



Reading: Exemplar: Using logs to help you track down an issue in Linux

Introduction

In this lab, you used logs to help you troubleshoot and track down an issue. As an IT Support Specialist, it's crucial that you know how to troubleshoot and "follow the cookie crumbs." There were five different issues that you resolved using the skills you learned so far in this course.

This exemplar is a walk-through of the previous Qwiklab activity, including detailed instructions and solutions. You may use this exemplar if you were unable to complete the lab and/or you need extra guidance in competing lab tasks. You may also refer to this exemplar to prepare for the graded quiz in this module.

What you'll do

In this lab, you will:

- Familiarize yourself with the process of changing permissions within a file and folder in Linux
- Change the ownership of a specific file and folder

Scenario

Your computer is having some issues, and you can't seem to figure out what's wrong. Argh! Dig through the logs to figure out how to fix these issues.

You'll use logs to identify issues on a Linux VM, which you'll then fix using the knowledge you've gained from the other labs that you've completed.

What you should already know

This lab focuses on looking at logs that indicate issues that need to be fixed. These issues can be resolved using the skills you've gained in previous labs, so detailed instructions won't be included here. You're on your own... but you've got this!

Here are the concepts you need to be familiar with before taking this lab:

- Updating software that's out-of-date
- Finding and deleting files
- Modifying file permissions
- Finding and terminating specific processes

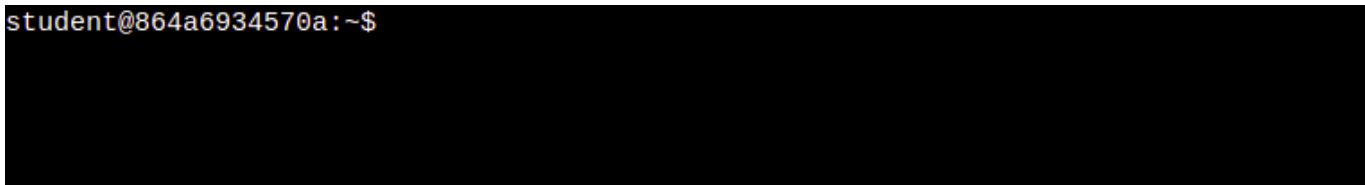
Start the lab

You'll need to start the lab before you can access the materials. To do this, click the green **Start Lab** button at the top of the screen.



Start Lab

After you click the **Start Lab** button, you will see a shell, where you will be performing further steps in the lab. You should have a shell like this:



```
student@864a6934570a:~$
```

Viewing logs on Linux

On Linux machines, logs are stored in the **/var/log** directory. There are lots of log files in this directory, and you can view them with this command:

```
ls /var/log
```

OUTPUT:

```
alternatives.log  auth.log  dpkg.log  faillog  messages  user.log
apt               btmp     exim4     lastlog  syslog    wtmp
```

We're interested in **syslog** for the moment. The logs on Linux can be viewed like any text file; you can use the command below to view the contents of syslog:

```
sudo cat /var/log/syslog
```

The log contents are super long, so you'll have to scroll through the logs to look for the five entries that are relevant to this lab. The logs are entered chronologically, and the logs that you'll need to fix should be timestamped around the time that the lab started. For convenience, all of the log entries you need to fix contain the phrase **Qwiklab Error**. Knowing this, you could also filter out the relevant logs using the **grep** command.

We'll walk through addressing one of the log's issues, then the other four will be up to you!

Low disk space!

Here's the log entry we will be dealing with first:

Aug 23 12:33:21 75b72110ff5a root: Qwiklab Error: Disk space is super low, fix it!

This error indicates that your computer is running out of memory due to a super large file. Unfortunately, it doesn't indicate which file is causing the problem, so you'll need to find it. Luckily, Linux has an easy way to find the largest files on your file system. The **du** command can be used to list all files in a directory (recursively through subdirectories, too), which you can sort by size to find the largest files. By piping the output of **du** (using the **|** symbol) to the sort command, you can sort the output by file size. The **-n** and **-r** flags tell sort to treat the string output on each line as a number (the file size), and to sort in reverse order so that the largest files are listed first. By piping the output of this into the **head** command, you can print out only the top few results (you can specify how many to output by adding **-n [NUMBER]** to the end of the command).

The command below uses **du**, **sort**, and **head** to show the top five largest files, starting from your **/home** directory:

```
sudo du -a /home | sort -n -r | head -n 5
```

OUTPUT:

```
3147112 /home
3147076 /home/lab
3145736 /home/lab/storage
3145732 /home/lab/storage/ultra_mega_large.txt
1328    /home/lab/downloads
```

You can see that the largest file in your home directory is **/home/lab/storage/ultra_mega_large.txt**, at about 5GB. This isn't an important file, but it's taking up a lot of space, so you can delete it to fix the disk space error:

```
sudo rm /home/lab/storage/ultra_mega_large.txt
```

Now that the large file is gone, this log's issue has been dealt with. You can see that the log entry is still present in the log file; logs aren't deleted once the errors that caused them are resolved.

The remaining log entries

Note: The following tasks in this lab require you to review tasks that you have completed in previous lab. You may refer back to those labs to review the steps.

The rest of the logs involve issues that you have already successfully fixed in earlier labs in this course. Refer back to those lessons and labs to refresh yourself on the required steps, if you're stuck:

Module 1 Lab: Finding and deleting files

In a previous lab, you found and deleted files. Review the lab in Module 1 titled Creating, Modifying and Removing Folders and Files in Linux to remove a corrupted file.

First, navigate to the file's location. It is in lab folder in the home directory. Next, use the `sudo` command **along with the command to remove files** to delete the file named `corrupted_file`.

Module 3 Lab: Software Packaging and File Archiving on Linux

In a previous lab you worked with software updates, file archiving, and more. Review the lab in module 3 titled Software Packaging and File Archiving on Linux to update VLC.

Before you get started on this step, go back to the home directory. Next, use the `sudo` command along with the `apt-get` package manager to identify and install missing dependencies that are required by other installed packages. Finally, use the command to display the status of a specific package. In this case, you will check the status of the **VLC media player** package.

Note: You can use the `-f` flag in your command to force the installation or update of a package, even if there are potential conflicts or errors.

Module 5 Lab: Maintain Efficient Processes in Linux

In a previous lab, you used commands to do some process maintenance on a Linux virtual machine. Review the lab in module 5 titled Maintain Efficient Processes in Linux to find and terminate a malicious process.

First, use the command used to list processes with `-aux` with the pipe symbol followed by the command that searches for text patterns within a file or stream of data and the process name `totally_not_malicious`. Next, kill the command using `sudo` and the process ID. Finally, use the command you started with again to make sure that the `totally_not_malicious` process has been ended.

Module 2 Lab: Creating, modifying, and removing file and folder permissions in Linux

In a previous lab, you learned how to manage user permissions on a Linux machine. You fixed permissions of some files and folders. Review the lab in module 2 to change the permissions of a secret file.

The file that you need to change the permissions to is titled `super_secret_file.txt`. Use the `sudo` command along with the command used to change permissions followed by the permission mode that grants read, write, and execute permissions.

Note: Please make sure that you are running the commands using **sudo**. The purpose of sudo is to execute the command given to it with root privileges.

Conclusion

Excellent job! You've successfully used logs to track down and fix issues on a Linux machine.
